

Machine Learning Basics

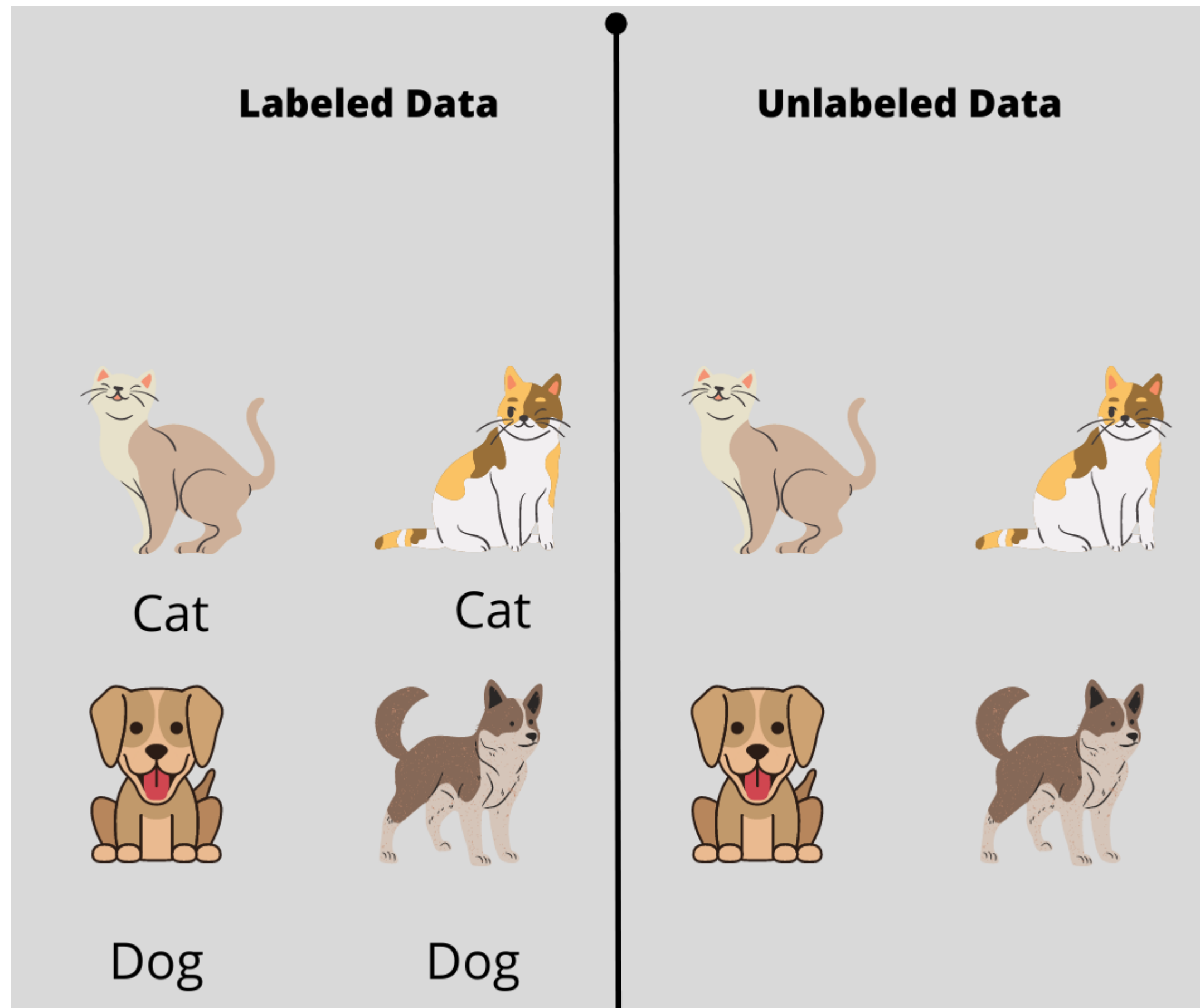
Defining labeled and unlabeled data

- **Labeled data:** Data with both input features and corresponding output labels. This is like having a teacher provide the answers alongside the questions.

Examples: Images of cats and dogs labeled as "cat" or "dog"; emails labeled as "spam" or "not spam"; customer reviews labeled with sentiment.

- **Unlabeled data:** Data with only input features and no output labels. This is like having a collection of images without knowing what they represent.

Examples: Customer transactions without fraud labels; text documents without topic labels; image datasets without content labels.



Supervised learning: Uses labeled data for predictive and classification models. This is widely used due to its practical value.

Examples: Image classification, spam detection, medical diagnosis.

Unsupervised learning: Uses unlabeled data to find patterns and structures without predefined labels. The model identifies patterns and relationships independently.

Examples: Clustering and dimensionality reduction.

The process of data labeling

- Data labeling assigns tags or descriptions to raw data, making it understandable to AI.
- It's often a collaboration between humans and AI, with human judgment being crucial for complex data.
- High-quality labeled data is vital for accurate and unbiased AI systems.

Why data labeling is important

- **Enables Machine Learning:** Labeled data helps machines understand data.
- **Improves Model Accuracy:** Well-labeled data enhances pattern recognition and prediction accuracy.
- **Mitigates Algorithmic Bias:** Labeling can reduce bias by ensuring diverse training datasets.
- **Facilitates Faster Model Training:** Provides curated datasets for more efficient training.
- **Supports Quality Assurance:** Essential for evaluating and validating models.

How to get labeled data

- **Manual labeling:** Humans annotate data points.
- **Automated labeling:** Uses rules or AI for automatic label generation.
- **Semi-automated labeling:** Combines human and automated methods.
- **Crowdsourcing:** Distributes tasks to a large group for efficiency.
- **Synthetic data generation:** Creates new data with labels from existing data or rules.
- **Open-source datasets:** Thousands of available datasets online.

Importance of using unlabeled data

- **Abundance:** Unlabeled data is vast and readily available from various sources.
- **Discovery of hidden patterns:** Can reveal correlations not seen with only labeled data.
- **Cost-effectiveness:** Avoids the time and expense of manual labeling.
- **Reduced human bias:** Allows models to find patterns without human interpretation.
- **Continuous learning:** Models can adapt to new trends by processing new data.

What Are Parametric Algorithms



Parametric Algorithms: Parametric algorithms operate under predefined assumptions regarding the distribution and relationships in the underlying data, utilizing a fixed number of parameters to predict new data points based on these assumptions.

Parametric Methods

- Parametric methods are statistical techniques that rely on specific assumptions about the underlying distribution of the population being studied.
- These methods typically assume that the data follows a known Probability distribution, such as the normal distribution, and estimate the parameters of this distribution using the available data.
- The basic idea behind the **Parametric method** is that there is a set of fixed parameters that are used to determine a probability model that is used in Machine Learning as well.
- Parametric methods are those methods for which we priory know that the population is normal, or if not then we can easily approximate it using a Normal Distribution which is possible by invoking the Central Limit Theorem.

Parameters for using the normal distribution methods are as follows:

- Mean

Population mean

Population mean formula	Explanation
$\bar{X} = \frac{\sum X}{N}$	• $\bar{X}$ = population mean • $\sum X$ = sum of each value in the population • N = number of values in the population

- Standard Deviation

The *population standard deviation* is the standard deviation of the entire population and often denoted by σ . It is given by the formula

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

where N is the size of the population consisting of $x_1, x_2, \dots, x_N$ and μ is the population mean.

Assumptions for Parametric Methods

Parametric methods require several assumptions about the data:

- **Normality:** The data follows a normal (Gaussian) distribution.
- **Homogeneity of variance:** The variance of the population is the same across all groups.
- **Independence:** Observations are independent of each other.

Machine Learning Models:

- **Linear regression:** Predicts a continuous outcome based on a linear relationship with one or more independent variables.
- **Logistic regression:** Predicts a binary outcome (e.g., yes/no) based on a set of independent variables.
- **Naive Bayes:** Classifies data points based on Bayes' theorem and assuming independence between features.
- **Hidden Markov Models:** Models sequential data with hidden states and observable outputs.

Advantages of Parametric Methods

- **More powerful:** When the assumptions are met, parametric tests are generally more powerful than non-parametric tests, meaning they are more likely to detect a real effect when it exists.
- **More efficient:** Parametric tests require smaller sample sizes than non-parametric tests to achieve the same level of power.
- **Provide estimates of population parameters:** Parametric methods provide estimates of the population mean, variance, and other parameters, which can be used for further analysis.

Disadvantages of Parametric Methods

- **Sensitive to assumptions:** If the assumptions of normality, homogeneity of variance, and independence are not met, parametric tests can be invalid and produce misleading results.
- **Limited flexibility:** Parametric methods are limited to the specific probability distribution they are based on.
- **May not capture complex relationships:** Parametric methods are not well-suited for capturing complex non-linear relationships between variables.

Applications of Parametric Methods

Parametric methods are widely used in various fields, including:

- **Biostatistics:** Comparing the effectiveness of different treatments.
- **Social sciences:** Investigating relationships between variables.
- **Finance:** Estimating risk and return of investments.
- **Engineering:** Analyzing the performance of systems.

What Are Non-Parametric Algorithms

Nonparametric Algorithms: Nonparametric algorithms do not make explicit assumptions about the data distribution or the predictor function, allowing for more flexibility and adaptability to the structure of the data, often at the expense of requiring more data and computational resources.



Nonparametric Methods

- Non-parametric methods are statistical techniques that do not rely on specific assumptions about the underlying distribution of the population being studied.
- These methods are often referred to as “distribution-free” methods because they make no assumptions about the shape of the distribution.
- In fact, the methods don't depend on the population. Here there is no fixed set of parameters available, and also there is no distribution (normal distribution, etc.) of any kind available for use.

Assumptions of Non-Parametric Methods

Non Parametric methods require several assumptions about the data:

- Independence: Data points are independent and not influenced by others.
- Random Sampling: Data represents a random sample from the population.
- Homogeneity of Measurement: Measurements are consistent across all data points.

Nonparametric Machine Learning Models:

- **K-Nearest Neighbors (KNN):** Classifies data points based on the k nearest neighbors.
- **Decision Trees:** Makes classifications based on a series of yes/no questions about the features.
- **Support Vector Machines (SVM):** Creates a decision boundary that maximizes the margin between different classes.
- **Neural networks:** Can be designed with specific architectures to handle non-parametric data, such as convolutional neural networks for image data and recurrent neural networks for sequential data.

Advantages of Non-Parametric Methods

- **Robust to outliers:** Non-parametric methods are not affected by outliers in the data, making them more reliable in situations where the data is noisy.
- **Widely applicable:** Non-parametric methods can be used with a variety of data types, including ordinal, nominal, and continuous data.
- **Easy to implement:** Non-parametric methods are often computationally simple and easy to implement, making them suitable for a wide range of users.

Disadvantages of Non-Parametric Methods

- **Less powerful:** When the assumptions of parametric methods are met, non-parametric tests are generally less powerful, meaning they are less likely to detect a real effect when it exists.
- **May require larger sample sizes:** Non-parametric tests may require larger sample sizes than parametric tests to achieve the same level of power.
- **Less information about the population:** Non-parametric methods provide less information about the population parameters than parametric methods.

Applications of Non-Parametric Methods

Non-parametric methods are widely used in various fields, including:

- **Medicine:** Comparing the effectiveness of different treatments.
- **Psychology:** Investigating relationships between variables.
- **Ecology:** Analyzing environmental data.
- **Computer science:** Developing machine learning algorithms.

Difference Between Parametric and Non-Parametric

Parametric Methods	Non-Parametric Methods
Parametric Methods uses a fixed number of parameters to build the model.	Non-Parametric Methods use the flexible number of parameters to build the model.
Parametric analysis is to test group means.	A non-parametric analysis is to test medians.
It is applicable only for variables.	It is applicable for both – Variable and Attribute.
It always considers strong assumptions about data.	It generally fewer assumptions about data.
Parametric Methods require lesser data than Non-Parametric Methods.	Non-Parametric Methods requires much more data than Parametric Methods.
Parametric methods assumed to be a normal distribution.	There is no assumed distribution in non-parametric methods.

Parametric data handles – Intervals data or ratio data.	But non-parametric methods handle original data.
Here when we use parametric methods then the result or outputs generated can be easily affected by outliers.	When we use non-parametric methods then the result or outputs generated cannot be seriously affected by outliers.
Parametric methods have more statistical power than Non-Parametric methods.	Non-parametric methods have less statistical power than Parametric methods.
As far as the computation is considered these methods are computationally faster than the Non-Parametric methods.	As far as the computation is considered these methods are computationally slower than the Parametric methods.
Examples: Logistic Regression, Naïve Bayes Model, etc.	Examples: KNN, Decision Tree Model, etc.

Parametric vs Nonparametric Algorithms Compared

Parametric Machine Learning Algorithms

Benefits

Simpler

Faster

Less Training Data

Limitations

Highly Constrained

Limited Complexity

Poor Fit

Nonparametric Machine Learning Algorithms

Benefits

High Flexibility

Power

High Performance

Limitations

More Training Data

Slower

Overfitting Training Data

Thank You!